

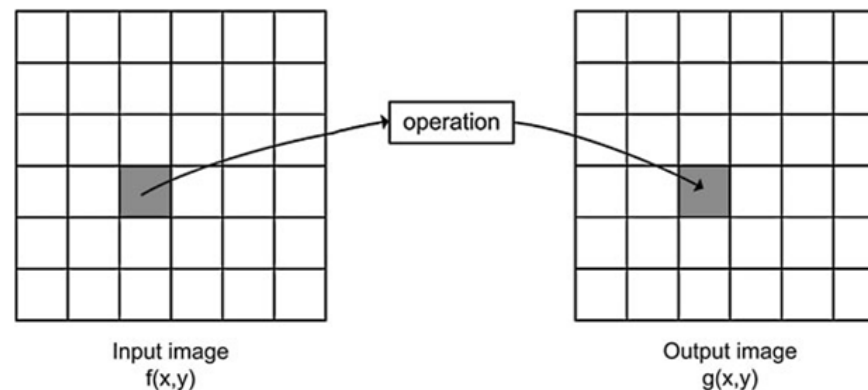
Biomedical Image Processing

POINT-TO-POINT TRANSFORMATIONS

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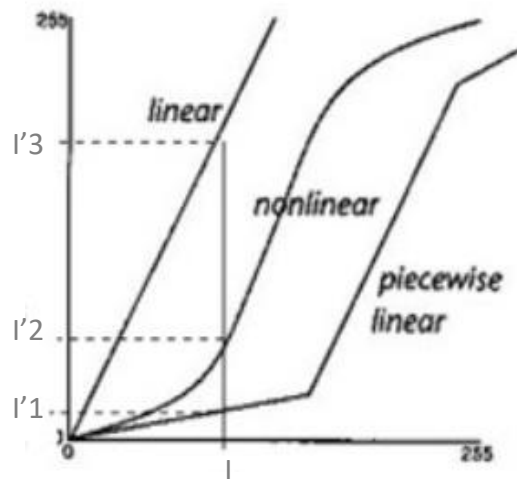
Point-to-point operations

- ❑ The simplest way to improve an image is through point-by-point processing;
- ❑ In these operations, each point of the resulting image $g(i,j)$ is obtained from a transformation T of the point with homologous coordinates of the initial image $f(i,j)$;
- ❑ It can be an operation of any type: arithmetic, logic, comparison,...

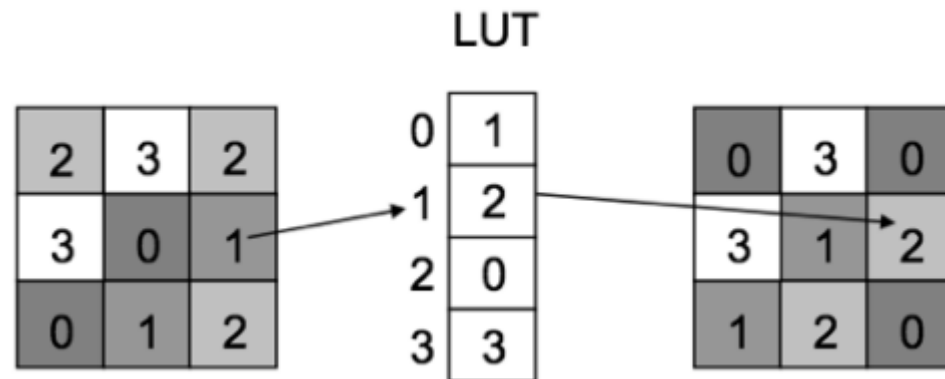


Point-to-point operations

- ❑ The transformation T must be a unique function whose domain is the range of allowed values for the original image;
- ❑ Known mathematical functions or “look-up-tables” can be used



Mathematical functions: $I' = T(I)$;



Look-up-tables

Brightness manipulation

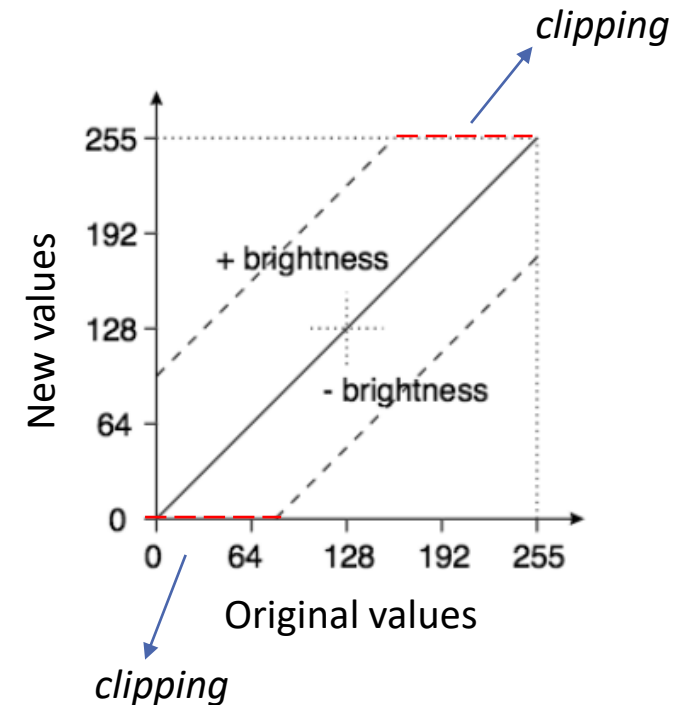
BRIGHTNESS

- ❑ Perception of image luminance: brighter or darker
 - ❑ Lower Intensities – darker image
 - ❑ Higher intensities – brighter image

BRIGHTNESS MANIPULATION

$$g(i,j) = f(i,j) + b;$$

- ❑ if $b < 0$, the image becomes darker;
- ❑ if $b > 0$, the image becomes brighter;
- ❑ We should clip the values that exceed the allowed dynamic range.



Brightness manipulation



Contrast manipulation

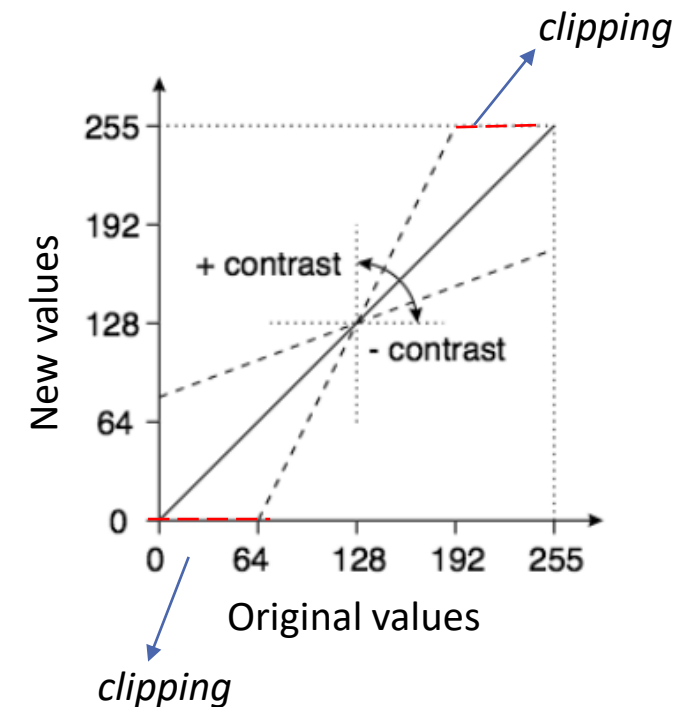
CONTRAST

- ❑ Difference between image luminances
 - ❑ High contrast – High difference between intensities
 - ❑ Less contrast – Less difference between intensities

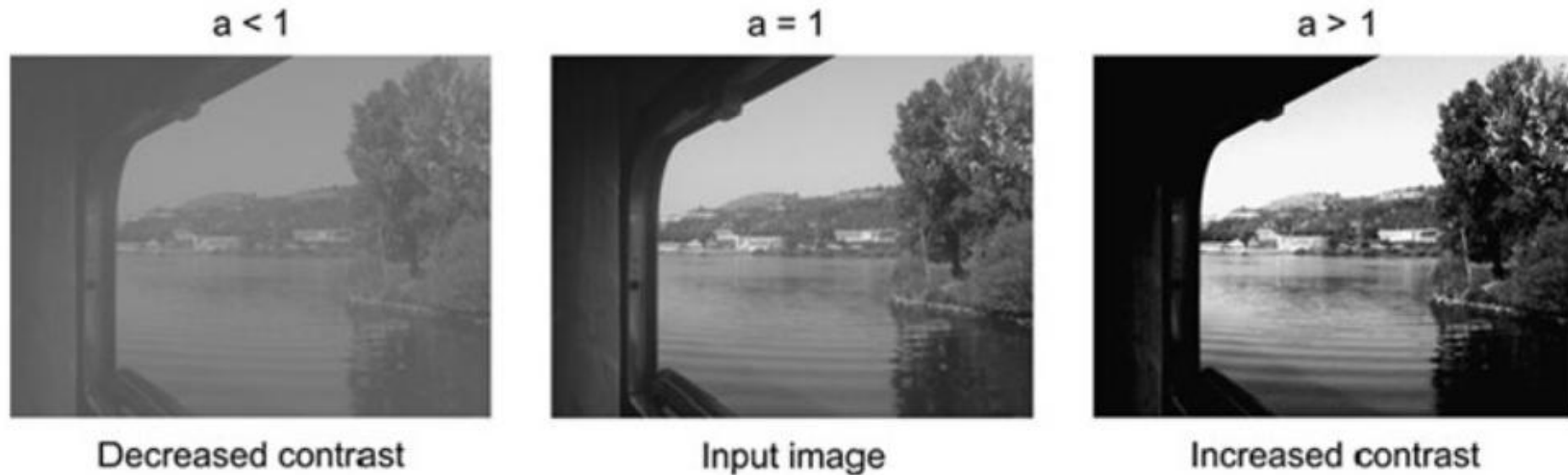
CONTRAST MANIPULATION

$$g(i,j) = f(i,j) \times \alpha + b;$$

- ❑ if $\alpha < 1$, the contrast decreases;
- ❑ if $\alpha > 1$, the contrast increases;
- ❑ We should clip the values that exceed the allowed dynamic range



Contrast manipulation



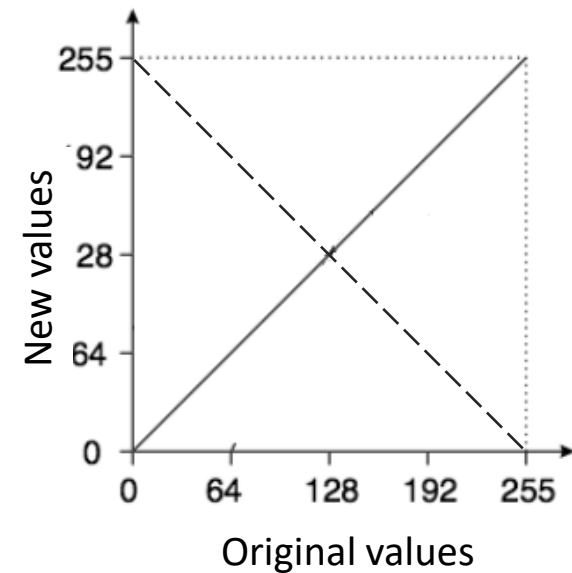
Inversion

INVERT AN IMAGE

- ❑ Process of replacing whites with blacks and blacks with whites;

$$g(i,j) = 255 - f(i,j);$$

- ❑ For an 8-bit image, consist of subtracting the value of $f(i,j)$ from 255;



Other transformations

NONLINEAR TRANSFORMATIONS

❑ Gamma;

$$g(i,j) = f(i,j)^{\gamma};$$

❑ Logarithmic;

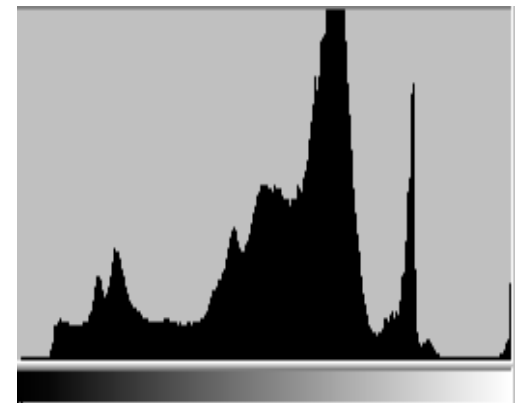
$$g(i,j) = c * \log(1+f(i,j));$$

❑ Exponential;

$$g(i,j) = c * (e^{f(i,j)} - 1);$$

Histograms

- ❑ Frequency of occurrence of image intensities
 - ❑ How many times does each intensity occur?
- ❑ Array that for each intensity value of an image contains an indication of the number of occurrences of that intensity
- ❑ Representation through a bar chart or graph
- ❑ Through the histogram, it is possible to know:
 - ❑ the contrast level of the image;
 - ❑ the average brightness of the image;
 - ❑ image exposure;

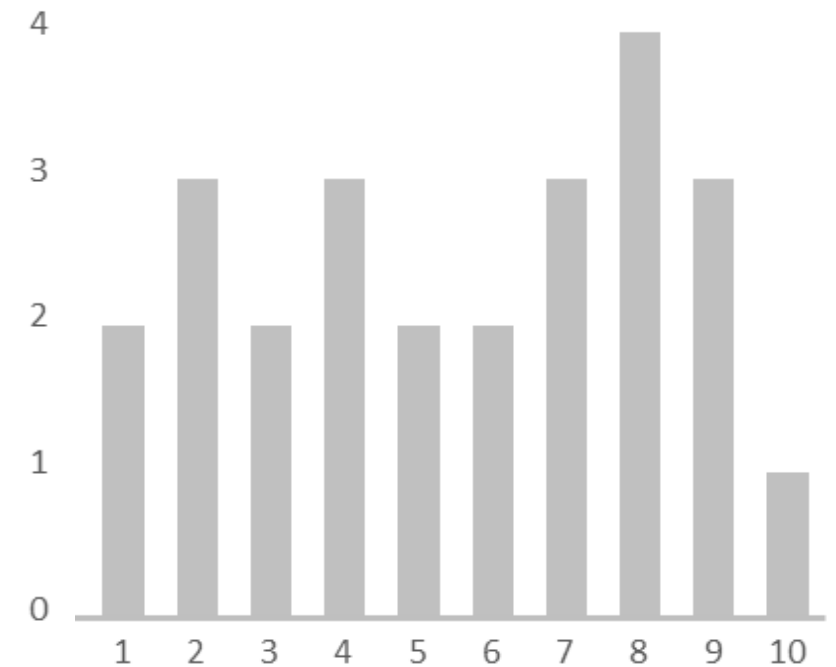


Histograms

□ Calculating a histogram

- For a given image $I(x,y)$ with:
 - N_i : number of occurrences of gray level i
 - G : number of gray levels

2	8	10	5	4
1	6	8	6	8
2	3	4	9	2
8	3	9	7	4
7	1	5	7	9



Histograms

EXERCISE

- Assuming the following image is 4-bit, draw its histogram

6	6	6	4	5
5	9	9	4	7
6	8	9	7	7
8	7	6	9	4
7	10	7	10	6

Histograms

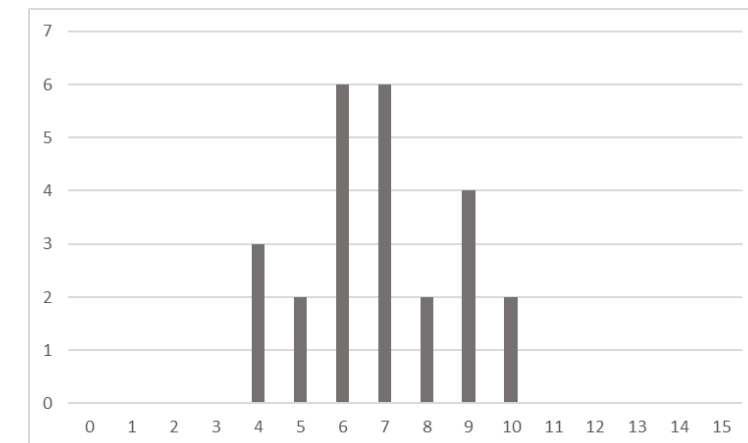
EXERCISE

- Assuming the following image is 4-bit, draw its histogram

Resolution

- 1) If the image is 4-bit, the histogram will be between 0 and 15;
- 2) Count the number of times each number is repeated in the image;
- 3) Build a bar chart
 - 1) Horizontal axis– 0 to 15 (4bits);
 - 2) Vertical axis– 0 to the count of the intensity that is repeated the most (mode)

6	6	6	4	5
5	9	9	4	7
6	8	9	7	7
8	7	6	9	4
7	10	7	10	6



Histograms

HISTOGRAM – ALGORITHM

- 1) For an image with G levels of gray, create an array h of dimension G and initialize it to zero;
- 2) Loop through each pixel and increment its position in the h array. If the pixel has intensity g then:

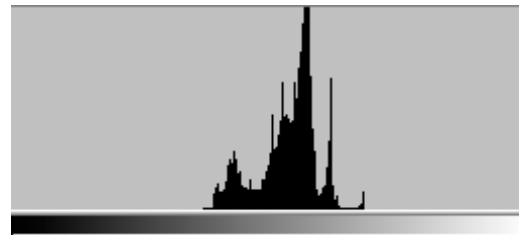
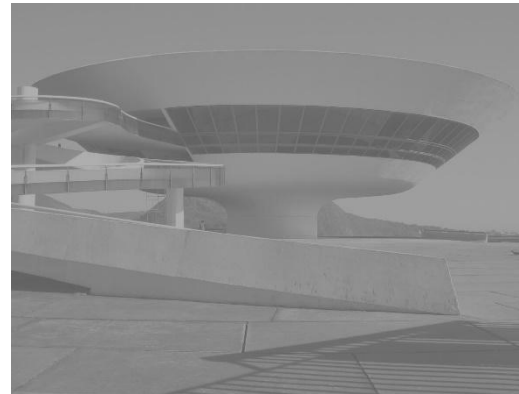
$$h[g] = h[g] + 1$$

Histograms

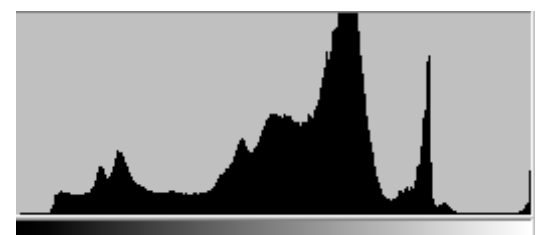
Image contrast

- Low contrast:
 - Histogram with concentrated intensity levels
- High contrast:
 - Histogram with scattered intensity levels

Low contrast



High contrast

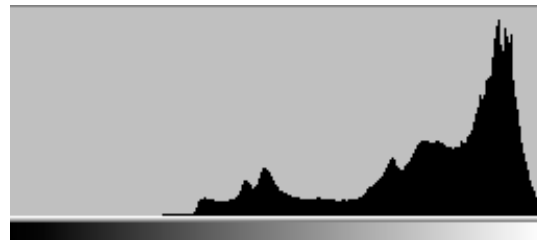


Histograms

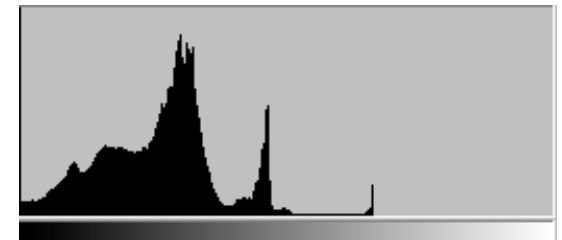
□ Image brightness

- High brightness:
 - Histogram with predominantly high-intensity levels
- Low brightness:
 - Histogram with predominantly low-intensity levels

High brightness



Low brightness

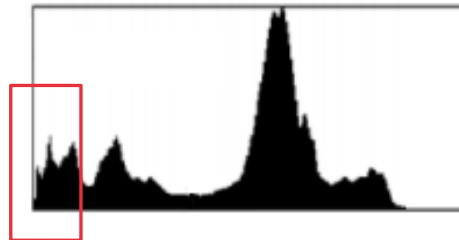


Histograms

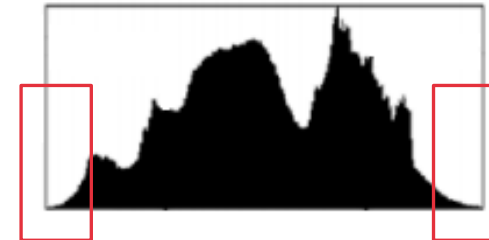
Image exposure

- Low exposure :
 - Histogram with high values of low intensities
- High exposure :
 - Histogram with high values of high intensities

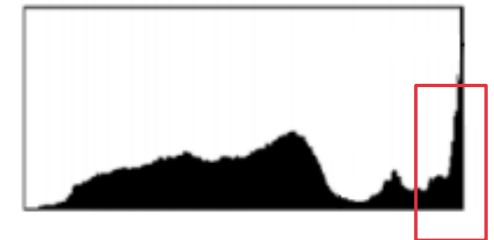
Low exposure



Good exposure



High exposure

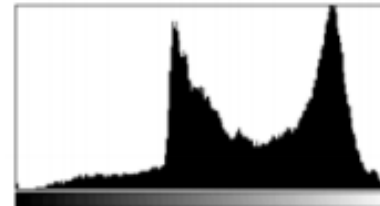


Histograms

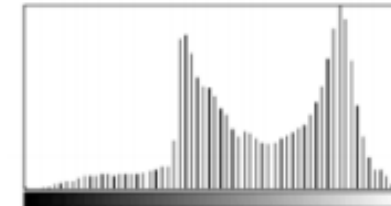
□ Dynamic range of the image

- High dynamic range:
 - Histogram with high number of intensity levels
- Low dynamic range:
 - Histogram with low number of intensity levels

High Dynamic Range



Low Dynamic Range

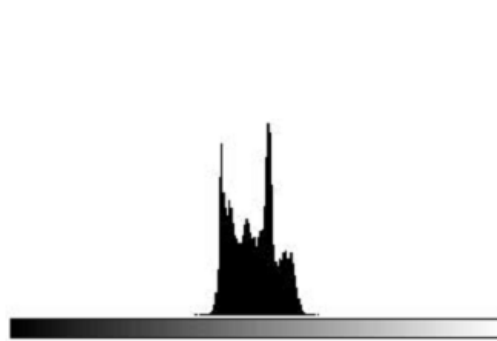


Histogram equalization

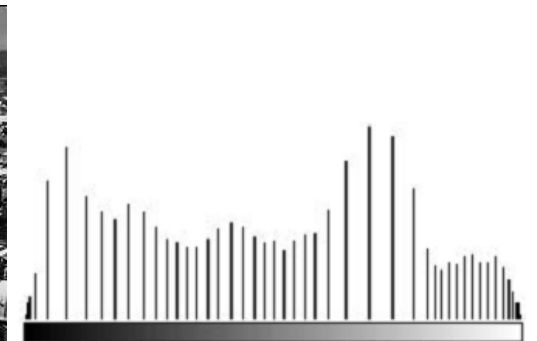
HISTOGRAM EQUALIZATION

- ❑ Redistributing the intensity values of an image to obtain a uniform distribution
- ❑ Point-to-point operation via a T transformation that uses the cumulative histogram
 - Intensity value of each pixel depends on its original value

Original Image



Equalized Image



Histogram equalization

CUMULATIVE HISTOGRAM

- The cumulative histogram is given by:

$$H[i] = \sum_{j=0}^i h[j], 0 \leq i \leq K$$

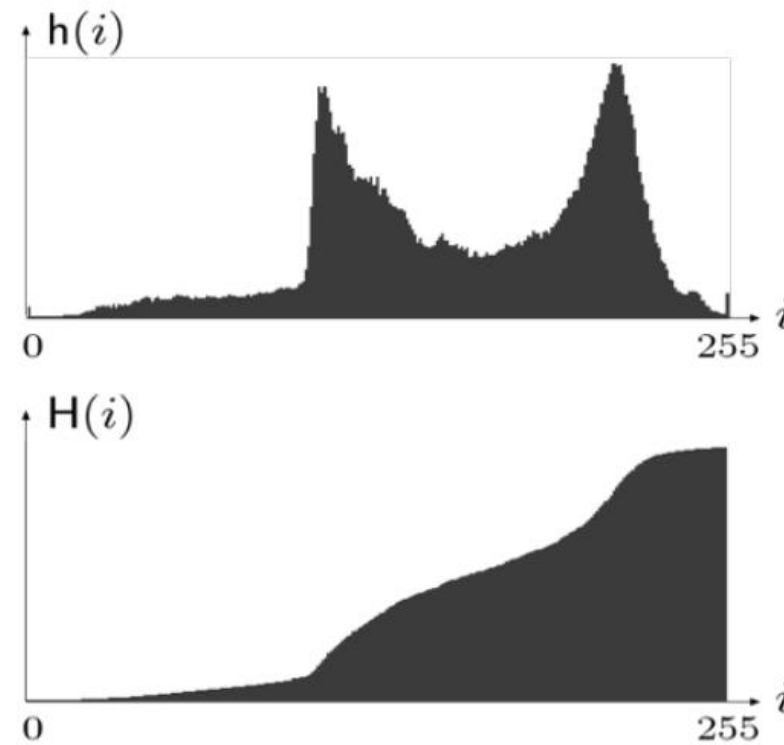
- K represents the maximum intensity
- We then have that a particular value of the cumulative histogram, $H[i]$, is the sum of all the values of the original histogram, $h[j]$, for $j \leq i$;
- The cumulative histogram is a monotonic increasing function, with its maximum value being:

$$H[K] = \sum_{j=0}^K h[j] = S$$

where $S = M \times N$ (M is the number of columns and N is the number of rows in the image)

Histogram equalization

CUMULATIVE HISTOGRAM



Histogram equalization

HISTOGRAM EQUALIZATION - ALGORITHM

- 1) For an image with S pixels with G levels of gray, create two arrays h and H of dimension G ;
- 2) Create the histogram h ;
- 3) Create the cumulative histogram H ;
- 4) Construct a table with the transformation law:

$$T[g] = \text{round}\left(\frac{G}{S}H[g]\right)$$

- 5) Analyse the input image and write the output image so that:
 $gs = T[g]$

Histogram equalization

EXERCISE

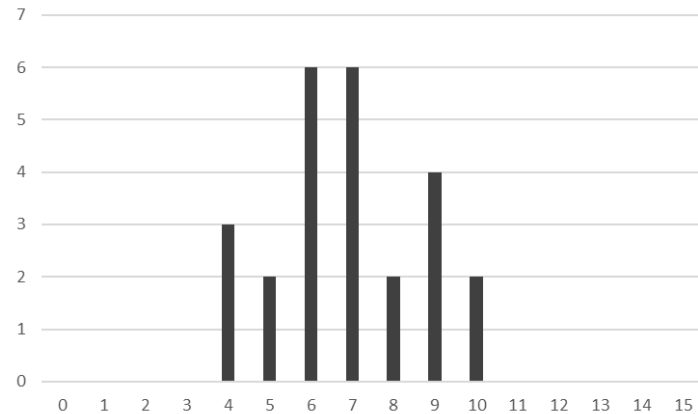
- Assuming that the next image is 4-bit, draw its histogram, its cumulative histogram and equalize the image. Show the histogram of the equalized image.

6	6	6	4	5
5	9	9	4	7
6	8	9	7	7
8	7	6	9	4
7	10	7	10	6

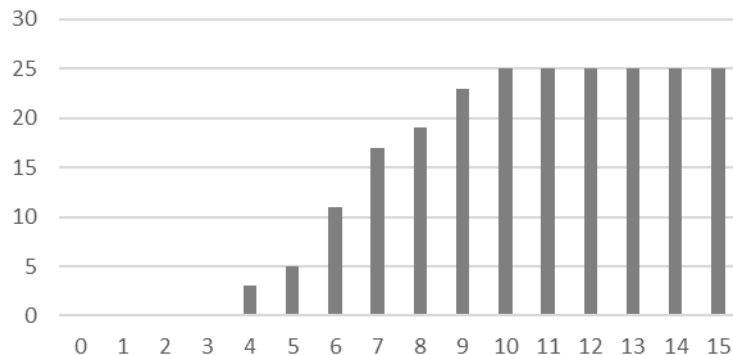
Histogram equalization

EXERCISE

Histogram



Histogram cumulative



Transform

$$T[g] = \text{round}\left(\frac{G}{S} H[g]\right)$$

- $g = 0$

$$T[0] = \text{round}\left(\frac{15}{25} \times H[0]\right) = \text{round}(0) = 0$$

...

- $g = 6$

$$T[6] = \text{round}\left(\frac{15}{25} \times H[6]\right) = \text{round}\left(\frac{15}{25} \times 11\right) = 7$$

...

- $g = 15$

$$T[15] = \text{round}\left(\frac{15}{25} \times H[15]\right) = \text{round}\left(\frac{15}{25} \times 25\right) = 15$$